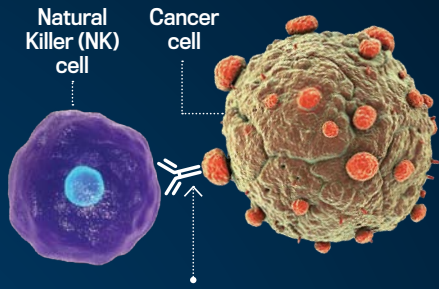
WHAT IS NK-CELL THERAPY?

It uses the NK or Natural Killer cells present in the human immune system. Unlike T cells, however, NK cells are not tailored to specific antigens

HOW IT IS ADMINISTERED



MONOCLONAL ANTIBODIES
NK therapy uses monoclonal antibodies to activate these cells, improve their stamina and longevity and direct them to eliminate cancerous cells in the body

PROGRESS SO FAR
Zumutor has developed ZM008, a monoclonal antibody that can target tumour cell receptors blocking NK cell inhibitory signals. This enables NK cell activation and an immune response to cancerous cells

COLD TUMOURS
Cold tumours have low accumulation of immune cells and, unlike hot tumours, respond poorly to drugs that inhibit proteins called checkpoints that often prevent white blood cells from attacking cancer cells. The new therapies help by improving immune cell infiltration

CANCERS IT CAN TREAT
Prostate and pancreas

COST
Rs 20-30 lakh in Europe. Price varies with the product and type of cancer

ALREADY AVAILABLE: 5 NEW CANCER TECHNOLOGIES YOU SHOULD KNOW OF

EASYCHECK-BREAST
Looks for circulating tumour cells in the peripheral blood. Made by Datar Genetics, it is available at all Apollo hospitals, for Rs 9,000. Helps detect early stage breast cancer

CYTOTRON
Uses rotational field quantum magnetic resonance to produce fast radio bursts that can induce programmed cell death in cancer cells and stop metastasis

LIQUID BIOPSY TEST
Analyses a low volume of blood to detect cancer cells. Offered by two companies—OncoDiscover and Strand. Useful for analysing information on tumours in sensitive locations

CERAVAC VACCINE
Developed by SII, the new vaccine is an easy and affordable way to prevent cervical cancer, which is caused by the Human Papillomavirus. Priced between Rs 200 and Rs 400 per dose

DOSTARLIMAB
Sold under the brand name Jemperli and marketed by GlaxoSmithKline (GSK), this immunotherapy drug blocks a certain protein in cancer cells that then helps the body fight off these diseased cells. Approved for endometrial cancer by FDA

years has concentrated on ways a human body's own immune system can be activated to recognise and destroy cancerous cells. As Dr Lalit Kumar, chairperson, oncology, Artemis Hospitals, Gurugram, explains, "The body has a mechanism to rid itself of disease. In cancer, somehow this mechanism gets stalled. A lot of the new treatments are about how to prime the body to dispense of the cells." Immunotherapy ensures only the tumour and not the healthy cells get targeted, causing far less side-effects. And should there be a recurrence, the immune system can again recognise the harmful cells.

"Cancer is a complex disease and involves multiple abnormalities at the level of DNA RNA, protein, metabolites, and so on," says Dr Sudha S. Murthy, clinical director at Datar Cancer Genetics, a molecular oncology facility in Nashik, Maharashtra. Several completed and ongoing studies in India as well as abroad have shown better clinical outcomes in patients who have received personalised anti-cancer treatment. They were selected on the basis of several parameters, among them an analysis of DNA, RNA and other biomarkers such as the number of circulating tumour cells, the amount of mutations found in tumour cells or the measure of certain proteins and genetic errors in cells.

India now has several notable ventures for immunotherapy, including Curadev, Bugworks Research, Immuneel Therapeutics, Zumutor and Aurigene Discovery Technologies. Immuneel Therapeutics announced its Series A fundraising round of $15 million last month. Earlier this year, Bugworks Research raised $18 million for the pre-clinical development of molecules that will target tumours where the compound adenosine, a chemical found in human cells, plays a role in neurotranmission. Zumutor plans to file an investigational new drug development with the US Food and Drug Administration for its NK-Cell Therapy molecule (similar to CAR-T therapy where the focus is on NK or 'natural killer' cells and not T cells) this year and recently raised $6.2 million in Series A4 funding.

"Most of the new solutions today," says